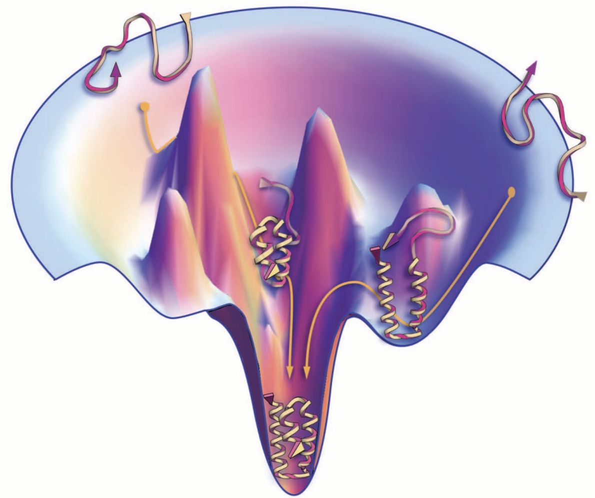
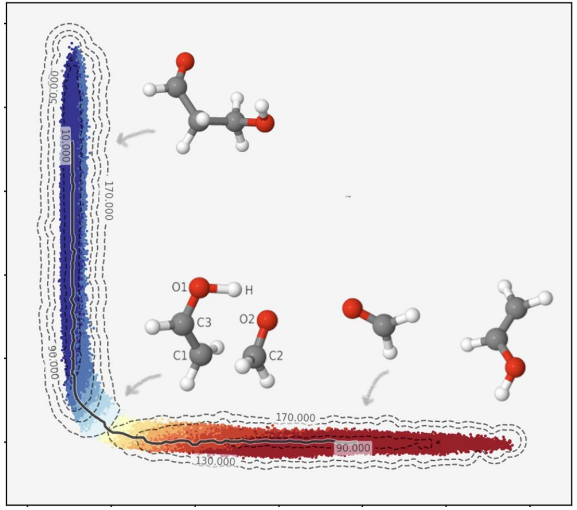
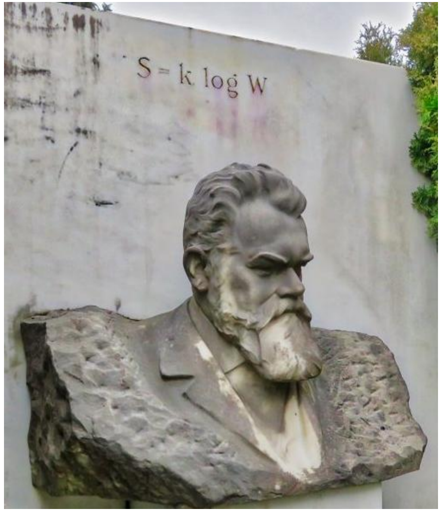
CH 412/512: Physical Chemistry: Statistical Mechanics and Kinetics







UNIVERSITY OF OREGON

Instructor: Dr. Dhiman Ray
Department of Chemistry and Biochemistry

Learning Objectives

1. Understand the basic principles of statistical mechanics: probability and partition function.

2. Employ these principles to calculate macroscopic thermodynamic properties, such as internal energy, free energy, heat capacity, etc.

3. Understand the basic principles of chemical kinetics and the temperature dependence of reaction rates.

Topics and Assessment

- Recapitulation of Thermodynamics
- Probability and Its Relation to Entropy
- Boltzmann Probability and Partition Function
- Molecular Partition Function
- Concept of Ensembles
- Chemical Kinetics

Weekly Homeworks (20%)

Midterm 1 (25%)

Midterm 2 (25%)

Final (30%)
(Covering all topics)

Instructors

Instructor

Dr. Dhiman Ray

Assistant Professor

Office Hours: Wed 2-3 pm, KLA 160E

Teaching Assistant

Revanth Elangovan

2nd Year Graduate Student

Office Hours: Thu 3-5 pm, ONY 384

Weekly Schedule

Monday, Tuesday, and Wednesday: Lecture by Dr. Ray (9.00 - 9.50 am)

Dr. Ray's Office Hours (Wed 2-3 pm. KLA 160E)

Revanth's Office Hours (Thu 3-5 pm, ONY 384)

Friday (9.00 - 9.50 am): Problem Solving Session by Revanth

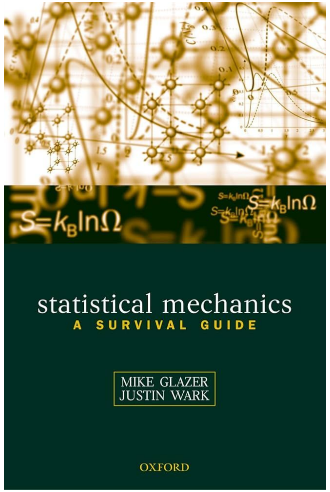
Weekly homework due at the beginning of the Friday class
Next homework posted sometime of Friday.

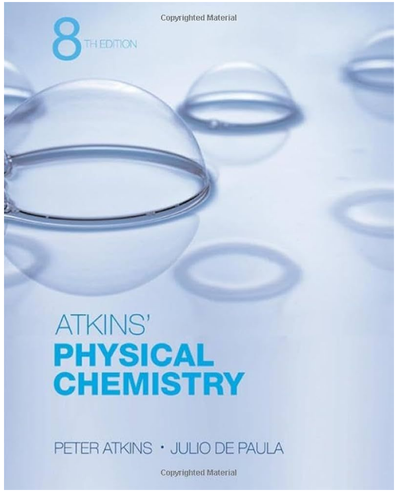
Problems in the Midterm and Final Exams will be very similar/identical to the ones discussed in Lectures, Problem Solving Sessions, and Homeworks.

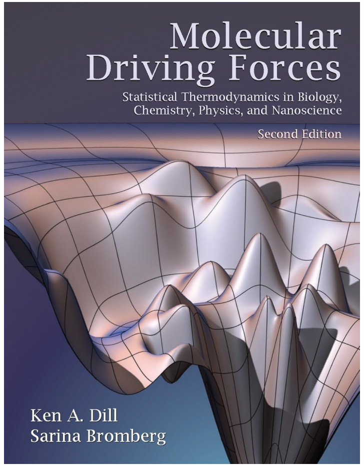
Textbooks and References

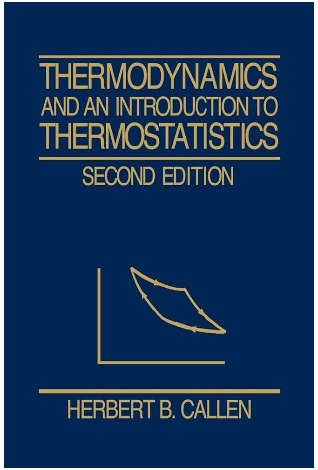
Teaching materials by Dr. Ray shared on Canvas

Additional References









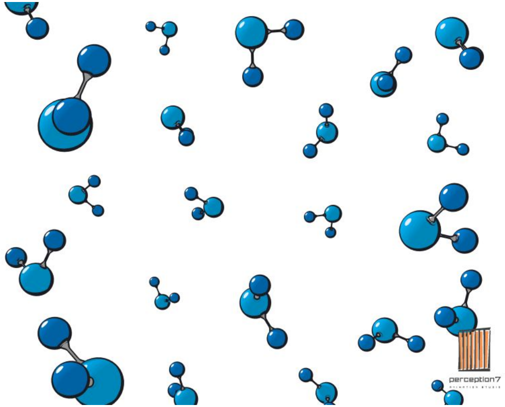
All necessary class notes and materials will be shared in Canvas

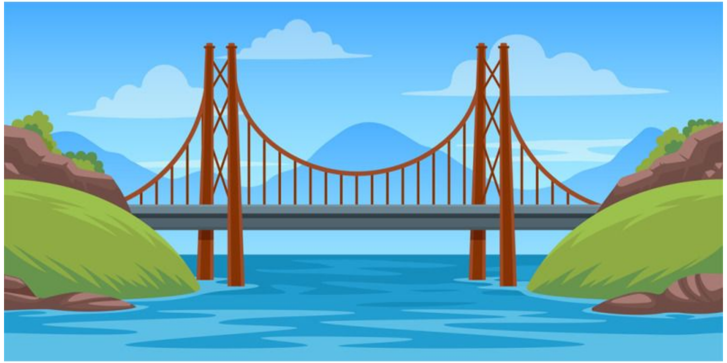
The Canvas Website

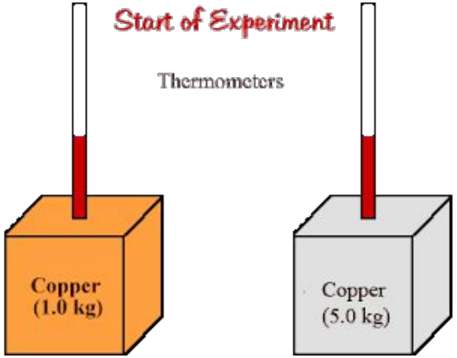
Introduction to Statistical Mechanics and Kinetics

What is Statistical Mechanics

Statistical Mechanics is the bridge between the microscopic and macroscopic worlds







Microscopic property:
Motion of individual atoms and molecules

Statistical Mechanics

Macroscopic property:
Thermodynamics e.g. heat capacity of materials

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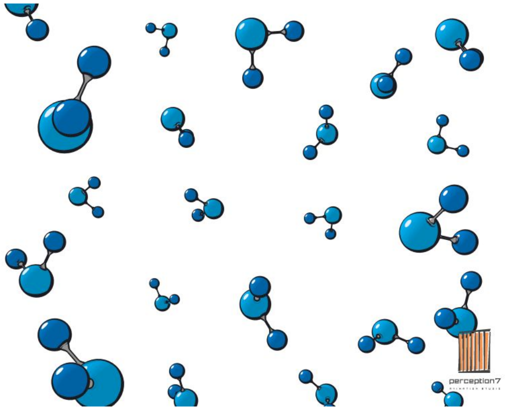
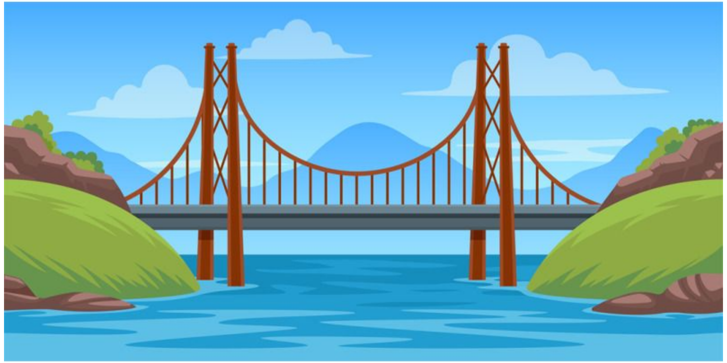
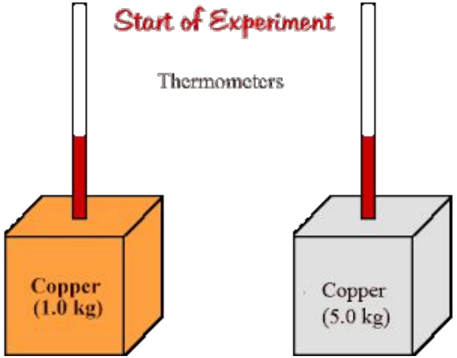






Quantum Mechanics
CH 413

Statistical Mechanics
CH 412

Thermodynamics
CH 411

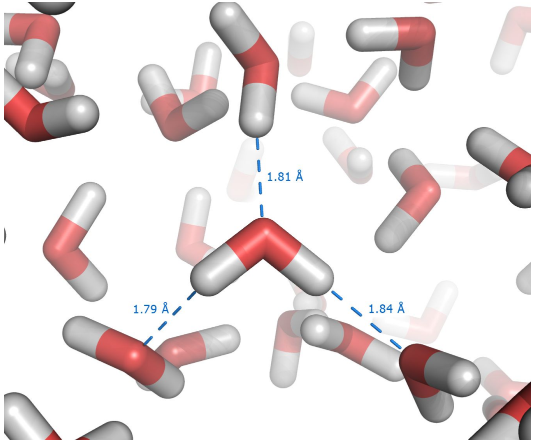
Microscopic: Statistical Mechanics

Macroscopic
Thermodynamics

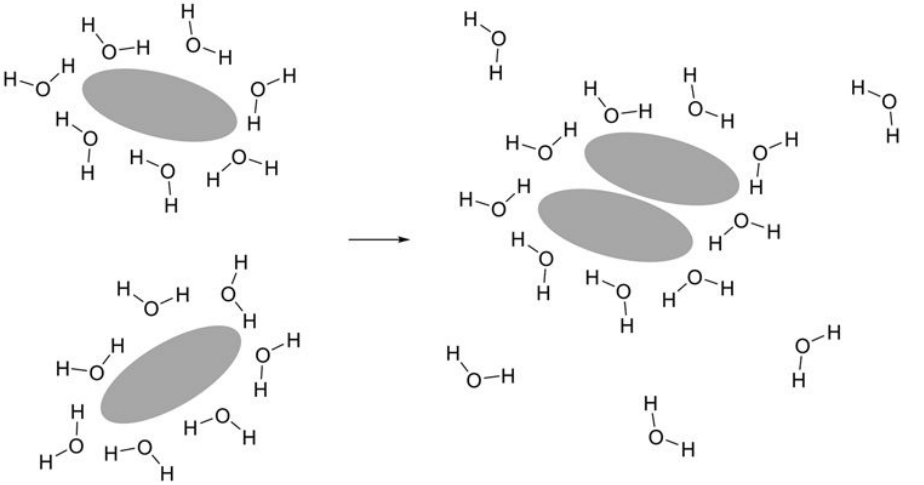
Martin Fitzner, Gabriele C. Sosso, Stephen J. Cox, and Angelos Michaelides
Journal of the American Chemical Society 2015 137 (42), 13658-13669

Hydrophobic Effect caused by water entropy

Water molecules can have any orientation in bulk



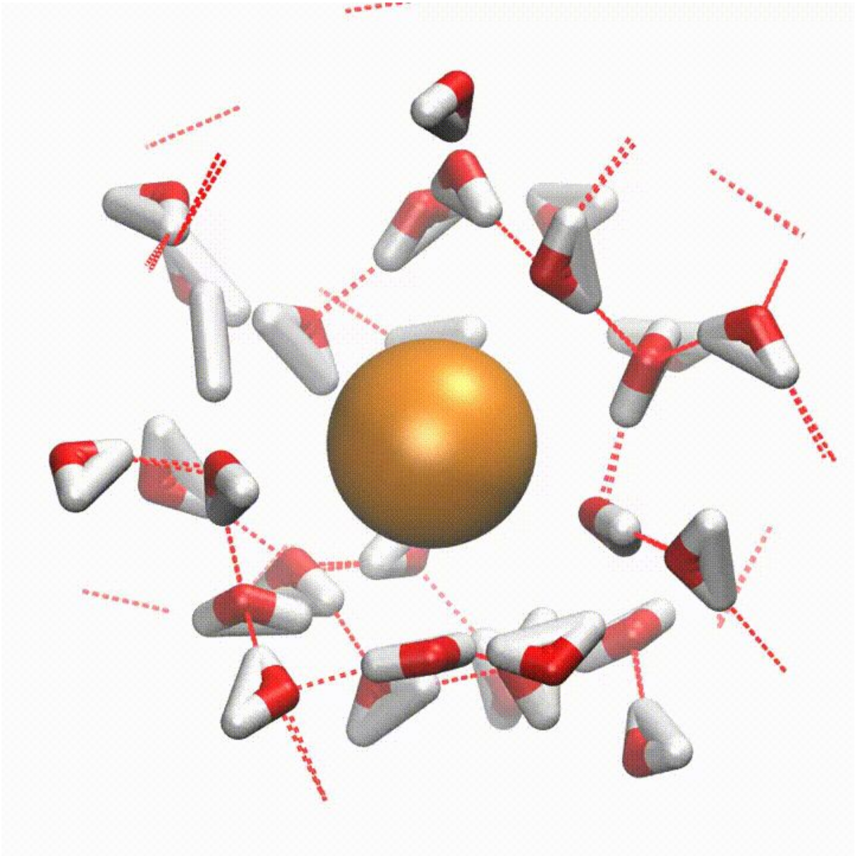
But not around hydrophobic molecules: they form H-bond with adjacent water molecules and form cage like structures



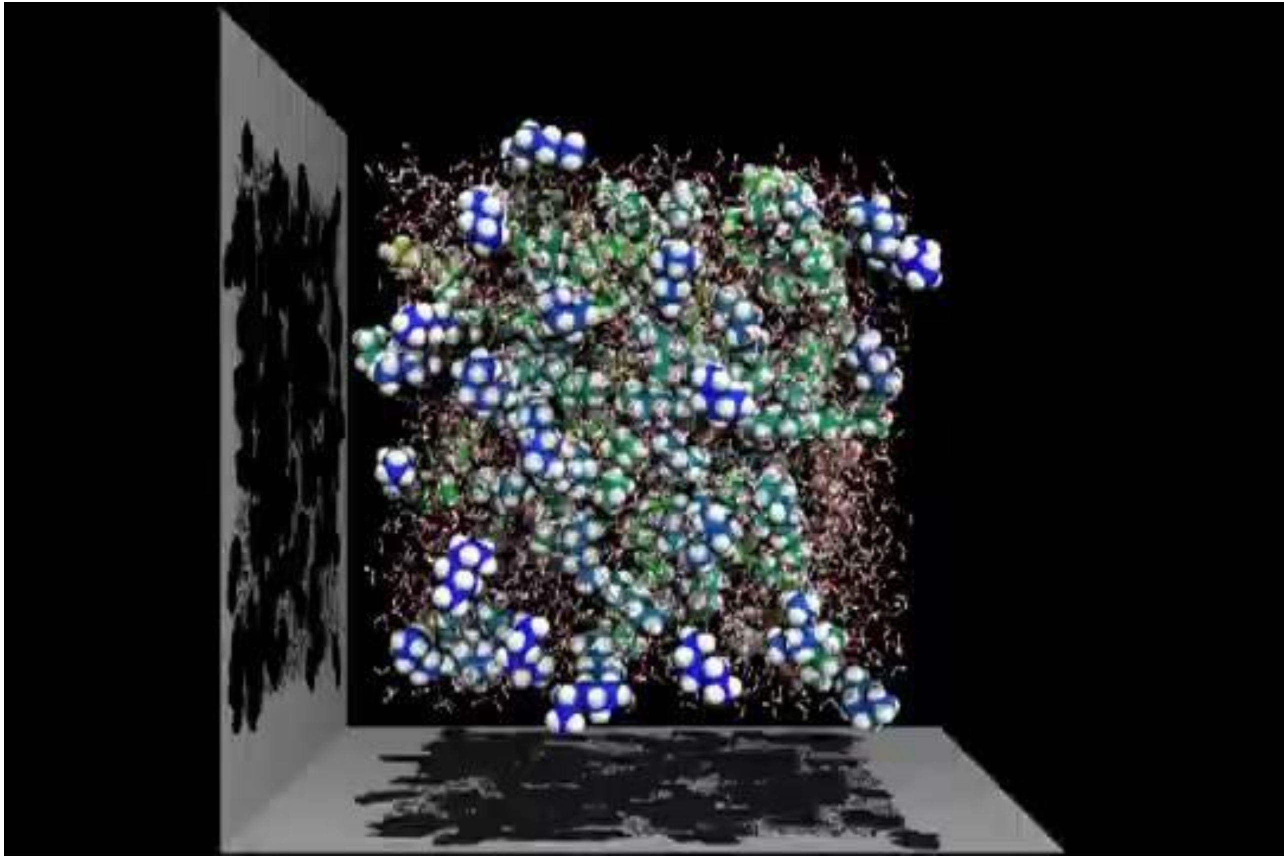
Less free water: low entropy

More free water: higher entropy

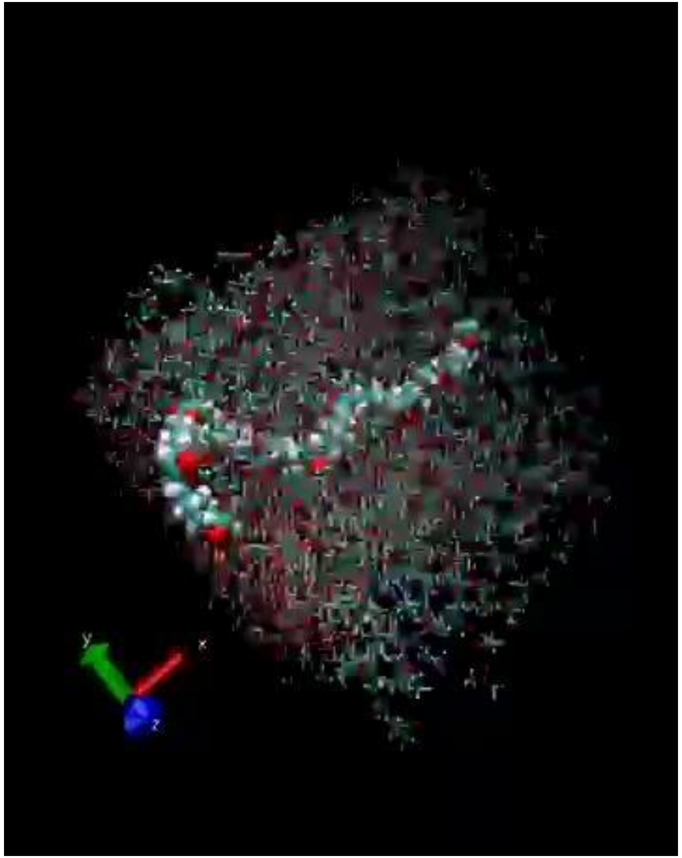
Hydrophobic Effect caused by water entropy



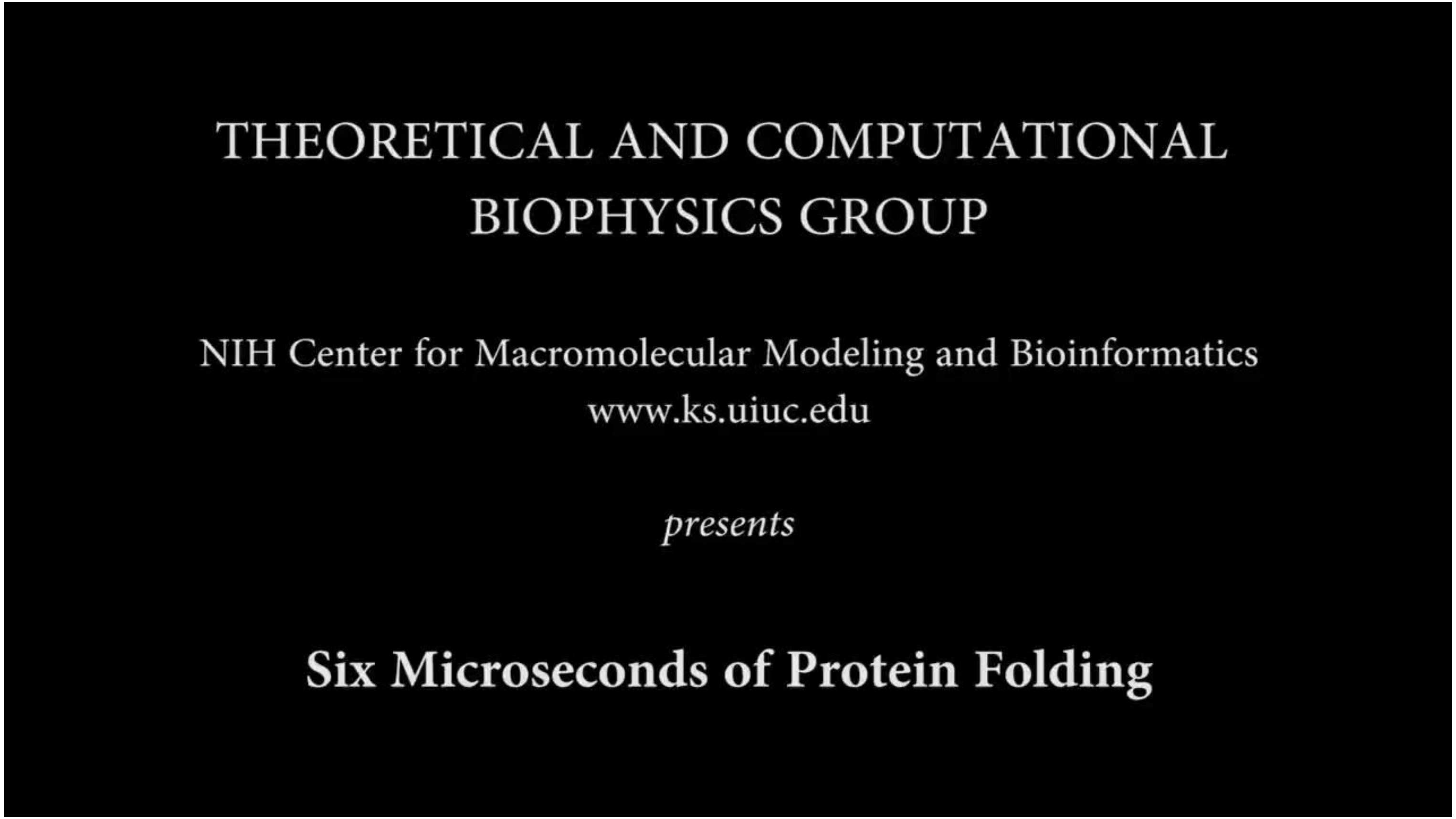
Hydrophobic Effect caused by water entropy



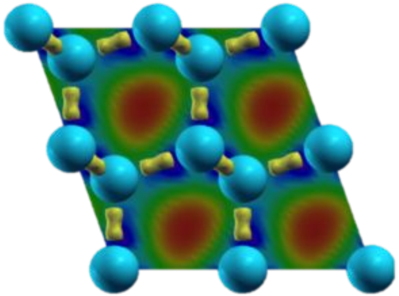
Hydrophobic Effect caused by water entropy



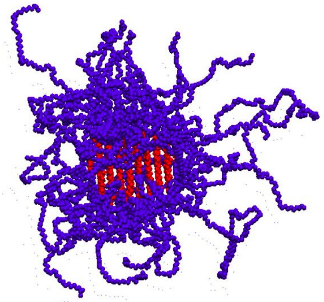
Protein folding simulation



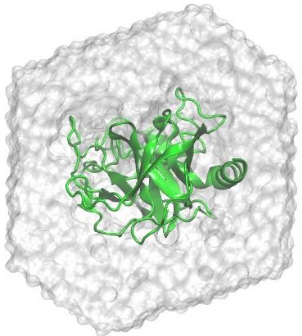
Where is statistical mechanics used?



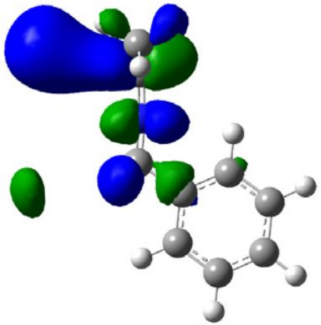
Modeling solid materials: (Solar cell, battery, superconductor etc.)



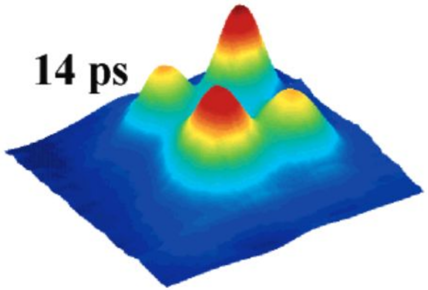
Polymer/soft material



Biomolecules (Study mechanisms, Drug design etc.)



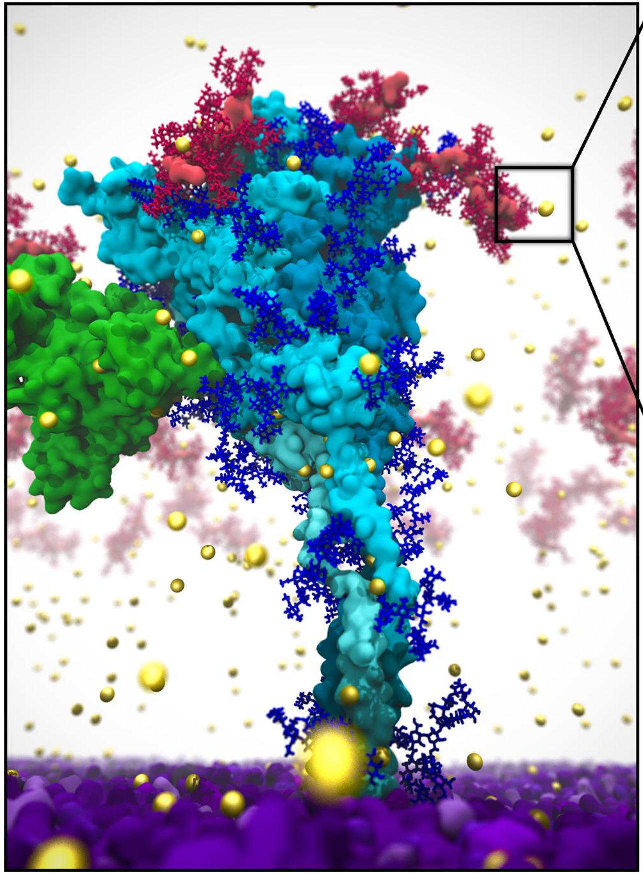
Organic/inorganic chemistry:
Rxn mechanism, catalyst design etc.

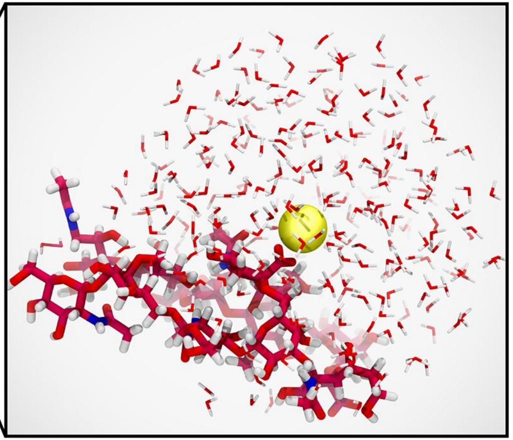


Spectroscopy: Signal analysis and interpretation

Figure courtesy: Gregory C. Rutledge, www.ce.studium.fau.eu, Liu et al. Angew Chem (2020), Zheng et al. (2006)

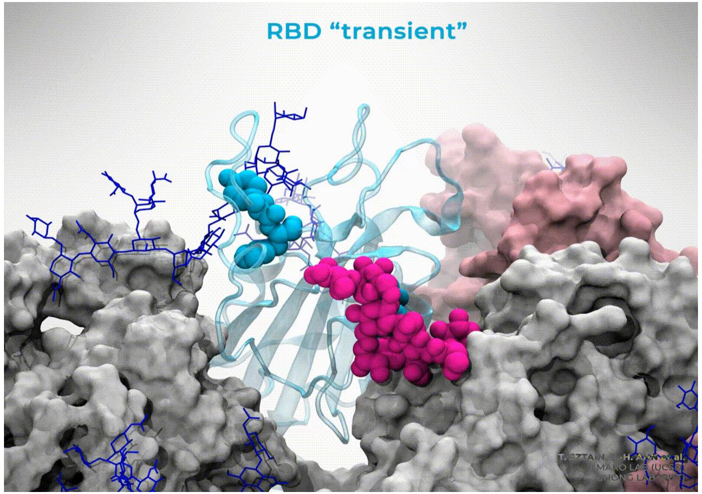
Some success stories: Understanding COVID-19





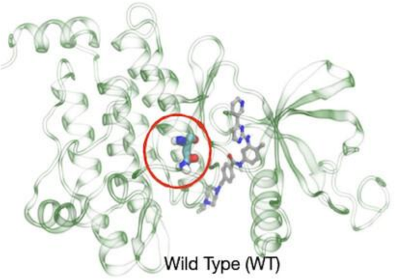
https://www.bnl.gov/newsroom/news.php?a=219440

Casalino, Lorenzo, et al. “AI-Driven Multiscale Simulations Illuminate Mechanisms of SARS-CoV-2 Spike Dynamics.” The International Journal of High Performance Computing Applications, vol. 35, no. 5, Sept. 2021, pp. 432–451

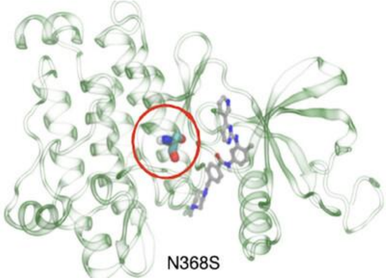


Sztain et al. *Nat. Chem.* 2021

Some success stories: Anticancer Drug Resistance



$$\Delta\Delta G = 0 \qquad \frac{k_{off}^{N368S}}{k_{off}^{WT}} \sim 3$$

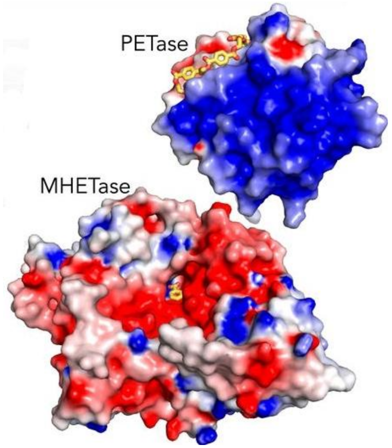


Kinetics determines how fast a reaction happens.

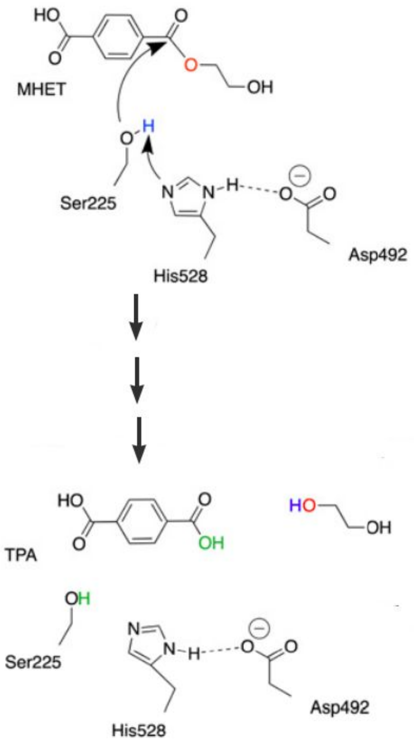
Blockbuster anti-cancer drug Gleevec became resistant due to N368S mutation.

Not because how strongly it binds (ΔΔG) but because how fast (k_{off}) the drug comes out of the cancer causing kinase protein.

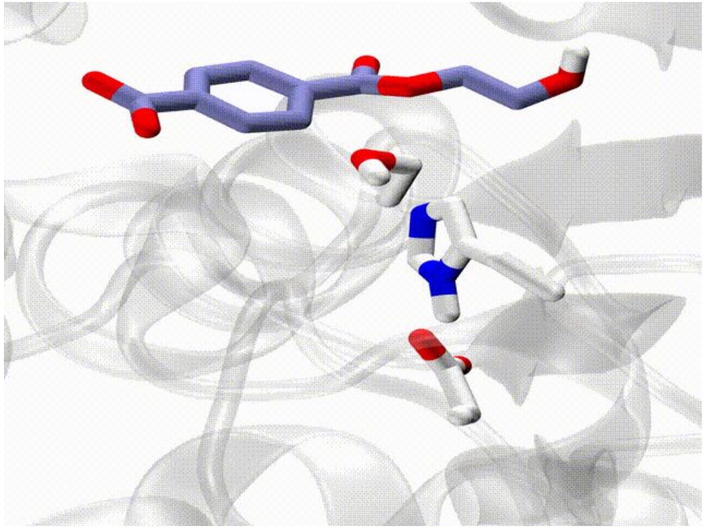
Some success stories: Plastic Degrading Enzyme



Ideonella sakaiensis



Knott et al. PNAS (2020)



From statistical mechanics we determine how thermodynamically stable are the reactant, product and the transition state.

From kinetics we determine how fast is the plastic degrading reaction.

Long Term Goals

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And many many more …..